

This Time Is Different? — Chapter 2 Blueprint

Literature Review: structure, arguments, sources, hypotheses and writing plan

Companion to the Research Plan v1.4 · 24 September 2026, updated 30 September 2026 · sample January 2003 – August 2023

How to use this: This blueprint tells you what each section must argue and which sources support it. It deliberately contains no ready-made paragraphs. Read the sources, form your own view and write every sentence yourself. ✓marks references whose details were confirmed online in this session; confirm all others in the DBS library before citing.

Updated 30 September 2026: (1) your results are in — use them to decide what 2.7 must set up, but keep Chapter 2 about the literature; (2) two exploratory extensions were added through Amendment A1 (Diebold–Yilmaz spillovers vs SHAP; the volatility paradox) — they get short literature support in 2.2 and 2.4 and are reported in Chapters 3–4; (3) correction: Gallagher & Twomey (1998), not Twomey (1998); (4) two new must-cite papers (Feng, Shi & Kutan, 2026; Brini, 2026); (5) the Literature Search Guide (28 Sep) adds about 45 papers by section — its IDs (A1...E5) are referred to below.

1. The whole dissertation at a glance

Word budgets assume about 15,000 words. Check the DBS handbook and scale every figure proportionally.

Chapter	Job it does	Main content	Words	Draft in
Abstract	Whole study in one paragraph	Question, data, methods, three key findings, contribution	300	Week 11
1 Introduction	Why this matters	Ireland as a small open economy hit by repeated crises; the forecasting and black-box problem; aim, RQ1–RQ4, contribution, structure	1,500	Week 11
2 Literature Review	What is known, what is not	Crisis theory → three crises & Ireland → transmission → volatility forecasting → explainable AI → gap and hypotheses	4,500–5,000	Weeks 1–5 (polish week 10)
3 Methodology	How, and why this way	Philosophy, CRISP-DM, data, target, alignment, models, walk-forward, tests, SHAP, ethics, AI-use declaration	3,500	Week 9
4 Results	What was found	Descriptives, RQ1 accuracy tables, RQ3 frozen models, RQ2 fingerprints, RQ4 stability, validation events, robustness; exploratory extensions (Amendment A1)	3,000	Week 9
5 Discussion	What it means	Findings against hypotheses and literature; history lens; implications for investors, regulators, researchers; what the exploratory checks add (two lenses on transmission; calm before crises)	2,000	Week 10
6 Conclusion	Answer and next steps	Answers to RQ1–RQ4, contribution, limitations, future research (the PhD extension)	1,000	Week 11
References & appendices	Evidence of rigour	Harvard references; code listing; data-quality report; extra tables	—	Throughout

2. What Chapter 2 must achieve

- **Show command of three fields:** financial crises, volatility econometrics, and explainable machine learning.
- **Build a theoretical lens:** why crises recur, and why their origin should shape how they spread.

- **Evaluate methods critically:** what GARCH, HAR, LSTM and SHAP can and cannot do.
- **Identify a precise gap:** not 'nobody has studied this', but exactly which combination is missing.
- **Derive the hypotheses:** by the end, RQ1–RQ4 and their hypotheses should look like the obvious next step.

The chapter works like a funnel, from broad theory to your exact question:

Step	Section	Question it answers
1	2.2 Why crises recur	Why should crises differ from each other at all?
2	2.3 Three crises and Ireland	How did each crisis start, and what happened in Ireland?
3	2.4 How shocks travel	How do we know a shock came from abroad, and where from?
4	2.5 Forecasting volatility	Which models forecast risk best, especially in crises?
5	2.6 Explaining ML forecasts	Can we see what drives a deep-learning forecast?
6	2.7 Gap and hypotheses	What is missing, and what exactly will this study test?

3. Search strategy (do this in weeks 0–2; summarise it in section 2.1)

- **Databases:** DBS library e-resources (check which of ScienceDirect, Wiley, Oxford Academic, JSTOR and EBSCO DBS subscribes to), Google Scholar, SSRN, RePEc/IDEAS, arXiv (preprints only, flagged).
- **Grey literature allowed:** central-bank and official reports (Honohan 2010; Regling & Watson 2010; Nyberg 2011) and major working-paper series (NBER, ECB, UCD).
- **Time range:** 1970 onwards for theory and classic methods; 2003–2026 for crisis evidence.
- **Inclusion:** peer-reviewed journals first; working papers only if widely cited or official; English.
- **Chaining:** for each key paper, check its reference list (backward) and 'cited by' in Google Scholar (forward). This is how you find the newest work.
- **Record-keeping:** Zotero with Harvard style; one note per paper using the synthesis matrix (section 6).

Theme	Search string (paste into Scholar / database)
Volatility + ML	("volatility forecasting" OR "realized volatility") AND (LSTM OR "neural network" OR "deep learning")
Transmission + Ireland	(contagion OR spillover* OR "volatility transmission") AND (Ireland OR Irish OR ISEQ OR "small open economy")
Explainability	(SHAP OR Shapley OR "explainable AI" OR "integrated gradients") AND (volatility OR finance OR "time series")
Irish crisis	("Irish banking crisis" OR "Irish sovereign" OR "bank guarantee") AND (equity OR stock OR contagion)
Other events	(COVID-19 OR Brexit OR "Ukraine") AND (contagion OR "volatility spillover")
Exploratory E1	("spillover index" OR connectedness OR "variance decomposition") AND (SHAP OR "explainable" OR "interpretable machine learning") AND volatility
Exploratory E2	("low volatility" OR "volatility paradox" OR "stability is destabilizing") AND ("banking crisis" OR "financial crisis")

4. Section-by-section blueprint

2.1 Introduction to the chapter (~250 words)

Purpose: Tell the reader what the chapter does, how the literature was found, and in what order it is reviewed.

Points to cover (ideas to argue in your own words):

- Scope: three crises (2007–09, 2010–12, 2020), the Irish market, 2003–2023.
- Search strategy in two or three sentences (databases, keywords, inclusion rules).
- Roadmap: one sentence per section 2.2–2.7.

Sources and how to use them:

Source	Use it to...
—	No citations needed beyond the search description.

Critical angle: None needed here; keep it short.

Feeds into: Signals rigour to the examiner before the content starts.

Avoid: *Starting with dictionary definitions or general statements about ‘the importance of finance’.*

2.2 Why financial crises recur: the theoretical lens (~700 words)

Purpose: Give the theories that make ‘this time is different?’ a real question, and show why crisis origin should matter for transmission.

Points to cover (ideas to argue in your own words):

- Efficient Market Hypothesis: returns should be close to unpredictable, but volatility clusters and is forecastable. So forecasting volatility is consistent with market efficiency. Make this point explicitly.
- Minsky’s Financial Instability Hypothesis: long calm periods encourage leverage and fragility (hedge → speculative → Ponzi finance). Ireland’s 2003–07 credit boom is a textbook case, and your 2003–07 training data is exactly that calm boom.
- The volatility paradox: Danielsson, Valenzuela & Zer (2018) turn Minsky’s idea into a measurable claim — across 60 countries and up to 211 years, prolonged below-trend volatility (not high volatility) preceded banking crises, through credit growth and leverage. One or two sentences here give your Minsky framing an empirical anchor; your exploratory E2 is reported in Chapter 4, not here.
- Kindleberger’s anatomy of crises (displacement, boom, euphoria, distress, panic).
- Reinhart & Rogoff: every boom is claimed to be ‘different’; crises recur and cluster; typology of banking, sovereign-debt and external crises.
- Synthesis: the theories disagree on predictability but agree that crises differ in origin and mechanism. That gives a testable expectation: transmission channels should differ by origin (RQ2), and models trained in calm periods should fail in crises (RQ3).

Sources and how to use them:

Source	Use it to...
Fama (1970) J. Finance 25(2)	state the efficiency baseline
Cont (2001) Quantitative Finance 1(2)	volatility clustering: why volatility is forecastable
Minsky (1986); Minsky (1992) Levy WP 74	stability breeds instability; the RQ3 logic
Kindleberger & Aliber (2015)	crisis anatomy; recurring pattern
Reinhart & Rogoff (2009)	title, typology, ‘this time is different’ syndrome
Kelly (2009) ✓	bridge to Ireland: Minsky dynamics in the Irish credit boom
Danielsson, Valenzuela & Zer (2018) RFS 31(7) ✓	the volatility paradox: evidence that calm precedes banking crises; basis of exploratory E2

Critical angle: These theories are mostly historical and qualitative. Few studies test them with forecasting models. Say that this is the opening your study uses.

Feeds into: RQ2 (origins differ), RQ3 (Minsky test).

Avoid: *Writing a history essay. Every theory paragraph must end with what it implies for your study.*

2.3 Three crises and the Irish market (~1000 words)

Purpose: Establish each crisis’s origin and mechanism, and what it meant for Ireland. This is where your history reading shows, and where each crisis-specific hypothesis gets its basis.

Points to cover (ideas to argue in your own words):

- 2.3.1 Ireland as a small open economy with an internationally integrated stock market, historically sensitive to UK and US shocks.

- 2.3.2 GFC (2007–09): US credit origin and global spread. In Ireland it met a home-grown property and credit bubble, the September 2008 bank guarantee, and a banking collapse. Key point from Bekaert et al.: contagion from the US was small, but contagion from domestic markets was large where fundamentals were weak. So expect both the US and the domestic channel to matter.
- 2.3.3 Irish sovereign debt crisis (2010–12): the bank–sovereign loop (bailouts raised sovereign risk, which weakened banks); euro-area dimension; troika programme (November 2010); strong Irish–European equity contagion that eased after intervention (Corbet & Twomey). So expect euro and domestic channels.
- 2.3.4 COVID-19 (2020): an exogenous shock hitting all markets at once; contagion concentrated in financial firms. So expect all three channels (domestic, US, euro) to rise together.
- 2.3.5 Validation events, briefly: the 2022 war and energy shock (euro channel) in the main model, and Brexit 2016 (UK origin) in the FTSE-for-DAX robustness run. Report the competing Brexit evidence honestly.
- Close with a comparison table: crisis × origin × mechanism × expected channel. It becomes the basis of H2 in section 2.7.

Sources and how to use them:

Source	Use it to...
Kearney (1998) J. Financial Research 21(1) ✓ partial	Irish market as a small integrated market; causes of Irish volatility
Gallagher & Twomey (1998) ESR 29(4) ✓	UK → Irish spillovers
Brunnermeier (2009) JEP 23(1)	GFC mechanism
Bekaert, Ehrmann, Fratzscher & Mehl (2014) JF 69(6) ✓	US vs domestic contagion in the GFC
Kelly (2009) ✓; Honohan (2010); Regling & Watson (2010); Nyberg (2011)	Irish crisis was substantially home-grown
Acharya, Drechsler & Schnabl (2014) JF 69(6) ✓	bank–sovereign loop; Ireland’s guarantee
Lane (2012) JEP 26(3); Whelan (2014) J. Macroeconomics 39(B) ✓	euro crisis; Irish crisis and recovery
Corbet & Twomey (2015) EFAJ 2015(3) ✓	Irish–European equity contagion, DCC-GARCH
Baker et al. (2020) RAPS 10(4); Akhtaruzzaman, Boubaker & Sensoy (2021) FRL 38 ✓	COVID shock and contagion
Li (2020) Economic Modelling 84 ✓	Brexit: UK influence on European markets fell during the campaign; co-volatility persisted after the vote
Boungou & Yatié (2022) Economics Letters 215 ✓	war shock hit world markets, strongest in the first two weeks
Tooze (2018)	narrative that links the GFC and the euro crisis

Critical angle: Most studies analyse one crisis at a time, with different methods and markets, so their findings cannot be compared directly. A single framework applied to several crises is missing (part of the gap). Note also that Li (2020) suggests the UK was not always the dominant transmitter around Brexit: your UK hypothesis could fail, and saying so in advance is a strength.

Feeds into: H2a–H2e (fingerprints), and why Ireland is a good laboratory.

Avoid: *Retelling events without analysis, and making claims about Ireland without an Irish source.*

2.4 How shocks travel: contagion, interdependence and volatility transmission (~700 words)

Purpose: Define the concepts precisely and review how transmission has been measured, ending with the limits of existing methods.

Points to cover (ideas to argue in your own words):

- Definitions: interdependence (normal linkages) vs contagion (a crisis-period increase beyond normal linkages). Correlations rise mechanically when volatility is high, which biases naive contagion tests.
- Classic evidence on volatility transmission between markets, including the role of trading hours (which also justifies your time-zone design).

- ‘Heat waves vs meteor showers’: is volatility driven by domestic news or by news travelling across markets? This maps directly onto your domestic vs foreign channels.
- Measurement toolkit: correlation and DCC-GARCH, BEKK, factor models, spillover indices (Diebold–Yilmaz), and recent deep-learning early-warning approaches.
- Spillover (connectedness) indices in more detail: Diebold & Yilmaz (2009) use VAR variance decompositions; their 2012 version uses a generalized decomposition (Pesaran & Shin, 1998) that does not depend on the order of the variables and adds directional and net measures; time-varying (TVP-VAR) versions are now common in Irish finance research. Applications that include Ireland: Search Guide B5 (Mensi et al., 2018), B7 (Aslam et al., 2021), B9 (Gołęb & Zamojska, 2019). This is the method of your exploratory E1.
- Synthesis: the methods measure co-movement, are mostly linear or pre-specified, and are not built to forecast. Results depend on the method chosen.

Sources and how to use them:

Source	Use it to...
Forbes & Rigobon (2002) JF 57(5)	the correlation bias; contagion vs interdependence
Pericoli & Sbracia (2003) J. Econ. Surveys 17(4)	definitions of contagion
King & Wadhwani (1990) RFS 3(1)	transmission of volatility between markets
Hamao, Masulis & Ng (1990) RFS 3(2) ✓	volatility spillovers across time zones
Engle, Ito & Lin (1990) Econometrica 58(3)	domestic (heat wave) vs foreign (meteor shower) volatility
Kearney (2000) Global Finance Journal 11 ✓	international volatility transmission
Bekaert, Harvey & Ng (2005) J. Business 78(1)	integration vs contagion (factor approach)
Dungey et al. (2005) Quantitative Finance 5(1)	review of contagion methods
Diebold & Yilmaz (2009; 2012)	spillover index; the 2012 set-up (daily range volatility, VAR(4), 10-day horizon, 200-day windows ✓full text) is the method of exploratory E1
Pesaran & Shin (1998) Economics Letters 58(1)	the generalized (order-invariant) variance decomposition behind DY (2012)
Antonakakis, Chatziantoniou & Gabauer (2020) JRFM 13(4)	TVP-VAR connectedness — name it as the PhD extension
Search Guide B5, B7, B9	DY applications that include Ireland (read before writing 2.4)
Sahiner (2023) Computational Economics ✓	closest prior work: spillovers across major crashes with econometrics plus deep learning

Critical angle: No single method is agreed; each captures a different notion of transmission. None uses the explanations of a forecasting model as evidence about channels. That is your methodological contribution. Planning note: your exploratory E1 shows the point on your own data — a VAR spillover measure and the SHAP shares agreed only weakly — so say which notion of transmission each measures.

Feeds into: RQ2; the time-zone alignment in Chapter 3.

Avoid: Using ‘contagion’ loosely. Define it once and use it consistently; call your SHAP result ‘transmission fingerprints’, not proof of contagion.

2.5 Forecasting volatility: from GARCH to deep learning (~900 words)

Purpose: Review the forecasting models and evaluation methods, and establish why the LSTM is worth testing against strong benchmarks.

Points to cover (ideas to argue in your own words):

- Stylised facts that make volatility forecastable (clustering, persistence, fat tails).
- ARCH and GARCH; why GARCH(1,1) is famously hard to beat.
- Realised variance and why the choice of proxy matters; range-based estimators (mention here, and explain your data-driven choice in Chapter 3).
- HAR: a simple long-memory model that is the standard realised-volatility benchmark.
- Evaluating forecasts when volatility is only measured with noise: QLIKE, Diebold–Mariano and the HLN correction.

- LSTM: why gated memory suits volatility; evidence in finance; hybrids with GARCH; evidence that LSTMs beat other networks and do well in high-volatility periods; evidence that ML is competitive with and can beat HAR.
- Correct validation for time series (no random shuffling).
- Synthesis: ML gains are real but modest and period-dependent; most evidence is on US indices; crisis-by-crisis evaluation in a small market is rare; the models are black boxes.

Sources and how to use them:

Source	Use it to...
Cont (2001)	stylised facts
Engle (1982); Bollerslev (1986)	ARCH and GARCH
Hansen & Lunde (2005) JAE 20(7)	GARCH(1,1) hard to beat: sets the bar
Andersen & Bollerslev (1998); Andersen et al. (2003)	realised variance; noisy proxies
Parkinson (1980); Garman & Klass (1980)	range estimators
Corsi (2009) JFEC 7(2)	HAR benchmark
Patton (2011) J. Econometrics 160(1) ✓	robust loss functions; QLIKE
Diebold & Mariano (1995); Harvey, Leybourne & Newbold (1997)	forecast comparison tests
Hochreiter & Schmidhuber (1997)	LSTM
Fischer & Krauss (2018) EJOR 270(2)	LSTM in financial markets
Kim & Won (2018) ESWA 103	LSTM–GARCH hybrids
Bucci (2020) JFEC 18(3) ✓	monthly S&P 500 data: LSTM and NARX best in MSE and QLIKE, also in 2007–09
Christensen, Siggaard & Veliyev (2023) JFEC 21(5) ✓	ML competitive with and beating HAR; gains grow with horizon
Bergmeir & Benítez (2012) Information Sciences 191	time-series validation
Search Guide C1–C3: Audrino & Chassot (2025) IJF; Kılıç (2025); Rahimikia & Poon	published evidence that HAR-type models often beat ML — company for your H1 result
Brini (2026) arXiv 2607.05291 ✓(preprint)	nine time-series foundation models vs HAR on 50 assets: no general advantage; an average of a small model and log-HAR most robust

Critical angle: Evidence is mixed: Hansen & Lunde point one way, Bucci and Christensen et al. the other. Present this as a genuine open question. It justifies RQ1 and makes a negative result publishable.

Feeds into: RQ1 and H1; RQ3 (crisis performance); the evaluation design in Chapter 3.

Avoid: *Explaining every model mathematically here. Put equations in Chapter 3; here, compare what studies found.*

2.6 Explaining machine-learning forecasts (~700 words)

Purpose: Introduce SHAP and its relatives, weigh their limits honestly, and show that explanations are rarely used to test economic mechanisms.

Points to cover (ideas to argue in your own words):

- Shapley values and SHAP: fair attribution of a prediction to inputs; properties (local accuracy, consistency).
- Gradient-based attribution for neural networks: Integrated Gradients and expected gradients (GradientSHAP). Explain why they suit an LSTM better than KernelSHAP.
- Limits: correlated inputs split credit arbitrarily; explanations can be unstable; explanation is not causation. Present the counter-argument that high-stakes decisions should use interpretable models, and note that your HAR and GARCH benchmarks are interpretable, so the study compares both worlds.
- Applications in finance: mostly feature rankings in credit and fraud models; in volatility, recent work uses SHAP descriptively (for example, identifying VIX and geopolitical risk as drivers).
- Synthesis: explanations are used to describe models, rarely as falsifiable evidence about how economic shocks are transmitted.

Sources and how to use them:

Source	Use it to...
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Lundberg & Lee (2017) NeurIPS	SHAP framework
Sundararajan, Taly & Yan (2017) ICML	Integrated Gradients
Erion et al. (2021) Nature Machine Intelligence 3	expected gradients = GradientSHAP
Aas, Jullum & Løland (2021) Artificial Intelligence 298	dependent (correlated) features
Kumar et al. (2020) ICML	problems with Shapley-based importance
Rudin (2019) Nature Machine Intelligence 1	the case against explaining black boxes
Molnar, Interpretable Machine Learning	clear background for your own understanding
Ahmed (2026) SSRN preprint ✓	recent LSTM-SHAP volatility work; flag as not peer-reviewed
Feng, Shi & Kutun (2026) JIMF 160, 103467 ✓	interpretable ML for cross-market volatility forecasting (VIX and non-VIX indices; US, Germany, Japan, Hong Kong, China, India; 2015–2024) — the closest recent work in spirit; must cite
Search Guide E1–E2: Choi, Shin & Shin (2024); Shin, Choi & Shin (2025)	vector SHAP grouped by region — the closest SHAP method; position your channel shares against it

Critical angle: Treat SHAP as a tool with known weaknesses. The response in your design (grouping, seed stability, validation events) comes directly from these criticisms. Say so; it shows the design is literature-led.

Feeds into: RQ2 and RQ4; the explainability design in Chapter 3.

Avoid: *Presenting SHAP as the ‘truth’ about the market.*

2.7 Synthesis, research gap and hypotheses (~500 words)

Purpose: Pull the strands together, state the gap precisely, and derive RQ1–RQ4 and the hypotheses.

Points to cover (ideas to argue in your own words):

- A short synthesis of what is known in each strand.
- The gap as a combination: (1) no common framework comparing transmission across crises of different origin for a small open economy; (2) forecasting studies rarely evaluate crisis by crisis or test whether models survive a new type of crisis; (3) explanations are rarely used as falsifiable evidence about transmission.
- RQ1–RQ4, each followed by its hypothesis and the literature it rests on (table in section 5 below).

Sources and how to use them:

Source	Use it to...
All sections above	each hypothesis must cite at least two sources

Critical angle: Be modest and precise. ‘Prior work has not combined X, Y and Z for Ireland’ is defensible; ‘this has never been studied’ is not.

Feeds into: Chapter 3 (how each hypothesis is tested).

Avoid: *A vague gap such as ‘few studies have looked at this’, or hypotheses that appear from nowhere.*

2.8 Chapter summary (~150 words)

Purpose: Three or four sentences: what was established and what Chapter 3 does next.

Points to cover (ideas to argue in your own words):

- What is known; what is missing; what this study will test.

Sources and how to use them:

Source	Use it to...
—	No new citations.

Critical angle: None.

Feeds into: Transition to Chapter 3.

Avoid: Introducing new material in the summary.

5. Hypothesis derivation table

Put a version of this table in section 2.7 (in your own words) and commit the hypotheses to Git before running any model.

Hypothesis	Literature basis	Expected result	What would contradict it
H1 (RQ1): LSTM beats GARCH and HAR out-of-sample (QLIKE)	For: Bucci (2020); Christensen et al. (2023). Against: Hansen & Lunde (2005)	Modest LSTM gain, larger in crises	DM-HLN not significant, or a benchmark wins
H2a GFC: US channel rises most; domestic also high	Brunnermeier (2009); Bekaert et al. (2014); Kelly (2009); Honohan (2010)	US and domestic shares above calm levels	Euro channel dominates
H2b Irish debt crisis: euro + domestic rise; US share below its GFC level	Lane (2012); Acharya et al. (2014); Corbet & Twomey (2015)	Euro/domestic shares highest	US share as high as in the GFC
H2c COVID: all three channels rise together	Baker et al. (2020); Akhtaruzzaman et al. (2021)	Broad, simultaneous increase	One channel dominates
H2d Brexit (FTSE robustness run): UK channel rises	Gallagher & Twomey (1998); Kearney (1998). Competing: Li (2020)	UK share above calm	No UK rise (consistent with Li 2020)
H2e 2022 shock: euro channel rises	Boungou & Yatié (2022)	Euro share above calm	US channel dominates
H3 (RQ3): frozen GFC model degrades most; Irish debt-crisis model least; COVID in between	Minsky (1992); Kindleberger & Aliber (2015); Reinhart & Rogoff (2009)	Ranking of QLIKE ratios as stated	Different ranking
H4 (RQ4): channel rankings stable across 5 seeds (mean Spearman ≥ 0.8)	Kumar et al. (2020); Aas et al. (2021)	Stable rankings	Rankings change with the seed

Exploratory extensions (Amendment A1, 30 Sep 2026). Not hypotheses and not part of 2.7: they were added after the results. Give them their literature basis briefly in Chapter 3 and report them in Chapter 4.

Extension	Literature basis	Expectation written in advance	Result that contradicts it
E1 SHAP vs Diebold–Yilmaz spillovers	Diebold & Yilmaz (2012); Pesaran & Shin (1998); Forbes & Rigobon (2002) on what co-movement measures mean	Descriptive comparison (8 sign checks, 5 rankings); no pass/fail	— (no test; disagreement is itself informative)
E2 volatility paradox	Danielsson, Valenzuela & Zer (2018); Minsky (1992); Hodrick & Prescott (1997)	Unusual calm (bottom 20%) before the GFC in Ireland and the US; not before COVID-19	No unusual calm before the GFC (what happened for Ireland and the US)

6. Synthesis matrix — build it as you read

Keep one row per paper in Excel or Zotero notes. Write each section of the chapter from the matrix, grouping rows by theme, never by author. The example rows use facts confirmed online in this session; check them when you read the papers.

Study	Market & period	Crisis	Method	Main finding	Relevance	Limitation
Corbet & Twomey (2015)	Irish & European equities, 2007–10 vs post-troika	Irish crisis	DCC-GARCH	Strong Irish–European contagion that eased after troika intervention	H2b; compare your euro-crisis channel	Correlation-based; no forecasting
Bekaert et al. (2014)	Country-industry portfolios, 55 countries	GFC	Factor model	Contagion from the US small; domestic contagion large where fundamentals weak	H2a domestic channel	Check data frequency when reading
Bucci (2020)	S&P 500, monthly,	incl. 2008	FNN, Elman,	LSTM and NARX best in	H1	US only;

	1950–2017		Jordan, LSTM vs ARFIMA	MSE and QLIKE, also in high volatility		monthly; no explanation
Li (2020)	European stock markets, 2016	Brexit	Time-varying multivariate GARCH; variance decomposition	UK influence fell during the campaign; co-volatility persisted after the vote	H2d competing evidence	Ireland not confirmed in sample
Diebold & Yilmaz (2012)	US stocks, bonds, FX, commodities; daily, Jan 1999 – Jan 2010	GFC	Generalized VAR variance decomposition; 200-day rolling	Spillovers limited until 2007; after Lehman large spillovers from stocks to other markets; full-sample total 12.6%	E1 method	Linear; US asset classes only
Danielsson, Valenzuela & Zer (2018)	60 countries, up to 211 years, monthly real returns	259 banking crises	One-sided HP trend; panel logit	Prolonged low volatility predicts crises (1% below trend for 5 years → +1.01% crisis probability); level of volatility does not	E2 method; Minsky anchor	Annual data; banking crises only
(your next paper)						

7. How to write it well

- **Paragraph pattern:** point → evidence from two or more sources → comparison or critique → what it means for this study. Every paragraph should earn its place by feeding a later hypothesis or design choice.
- **Synthesise, don't list:** a common weakness in dissertation drafts is 'Author A found... Author B found...'. Instead group studies: 'Studies using correlation-based tests (A; B) find..., whereas factor-model studies (C) find..., because...'.
- **Useful verbs:** extends, contradicts, qualifies, is consistent with, is limited by, overlooks, reconciles.
- **Primary sources for history:** use the official Irish reports and journal articles, not newspapers, for claims about the crisis. Newspapers are acceptable only for recent facts (such as the 2023–24 ISEQ exits).
- **Preprints and working papers:** cite them, but label them (e.g. 'SSRN preprint') and never rest a key claim on one alone.
- **Tense:** past tense for what a study did and found; present tense for what is generally accepted.
- **A summary table of prior studies** (like the matrix above, condensed) at the end of 2.4 or 2.5 is valued by examiners.
- **Harvard referencing** throughout, generated from Zotero; check every entry against the source.
- **Your own voice:** the critical comments and the links to your study are where marks come from. Those must be your own reasoning.

8. Reading and writing schedule for Chapter 2

Week	Read (and add to the matrix)	Write
0–1	Reinhart & Rogoff (ch.1); Minsky (1992); Fama (1970); Cont (2001); Kelly (2009)	Notes for 2.2
1–2	Honohan (2010); Lane (2012); Whelan (2014); Acharya et al. (2014); Bekaert et al. (2014); Corbet & Twomey (2015)	Notes for 2.3; crisis comparison table
2–3	Forbes & Rigobon (2002); Hamao et al. (1990); Engle, Ito & Lin (1990); Kearney (1998, 2000); Sahiner (2023)	Notes for 2.4
3–4	Hansen & Lunde (2005); Corsi (2009); Patton (2011); Bucci (2020); Christensen et al. (2023)	Notes for 2.5
4–5	Lundberg & Lee (2017); Erion et al. (2021); Aas et al. (2021); Kumar et al. (2020); Rudin (2019)	Draft 2.2–2.4
6–8	COVID, Brexit and 2022 papers; anything found by chaining	Draft 2.5–2.7
10	—	Polish the whole chapter; update the gap with your actual design
Now (Oct)	Diebold & Yilmaz (2012) pp.57–61; Pesaran & Shin (1998); Danielsson et al. (2018) pp.3–6, 13–14, 19–22 (accepted version); Feng et al. (2026); Brini	Method notes for the exploratory section of Chapter 3

	(2026)	
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9. Reference list for Chapter 2 (grouped by section)

✓= confirmed online in this session. Unmarked = standard references; confirm volume and pages before citing.

2.2 Theory

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Exploratory extensions (Chapter 3; not Chapter 2)

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